

AIM : Key Exchange using Diffie-Hellman

Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CLI Version :

Code :

```
# Diffie-Hellman Key Exchange
# Cyber & Information Security - MU NEP 2020 Sem 5
```

```
print("=" * 55)
print("    DIFFIE-HELLMAN KEY EXCHANGE")
print("=" * 55)
```

```
# Public values
p = int(input("\nEnter prime number (p): "))
g = int(input("Enter primitive root (g): "))
```

```
# Private keys
a = int(input("\nEnter private key of Alice: "))
b = int(input("Enter private key of Bob: "))
```

```
# Alice generates public key
A = pow(g, a, p)
```

```
# Bob generates public key
B = pow(g, b, p)
```

```
# Shared secret keys
alice_secret = pow(B, a, p)
bob_secret = pow(A, b, p)
```

```
print("\n" + "-" * 55)
print("PUBLIC PARAMETERS")
print("-" * 55)
```

```
print("Prime (p):", p)
print("Primitive Root (g):", g)
```

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```
print("\n" + "-" * 55)
print("KEY EXCHANGE")
print("-" * 55)

print("Alice Private Key:", a)
print("Alice Public Key :", A)

print("\nBob Private Key :", b)
print("Bob Public Key  :", B)

print("\n" + "-" * 55)
print("SHARED SECRET")
print("-" * 55)

print("Alice's Shared Secret:", alice_secret)
print("Bob's Shared Secret :", bob_secret)

if alice_secret == bob_secret:
    print("\n✓ KEY EXCHANGE SUCCESSFUL")
    print("Both Alice and Bob have the same secret key.")
else:
    print("\n✗ KEY EXCHANGE FAILED")

print("=" * 55)
```

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Output :

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mvluc>python "C:\Users\mvluc\Desktop\T085 Cyber Security Practical\Practical 5 CLI.py"

=====
DIFFIE-HELLMAN KEY EXCHANGE
=====

Enter prime number (p): 31
Enter primitive root (g): 7
Enter private key of Alice: 3
Enter private key of Bob: 5

-----
PUBLIC PARAMETERS
-----
Prime (p): 31
Primitive Root (g): 7

-----
KEY EXCHANGE
-----
Alice Private Key: 3
Alice Public Key : 2
Bob Private Key : 5
Bob Public Key : 5

-----
SHARED SECRET
-----
Alice's Shared Secret: 1
Bob's Shared Secret : 1

✓ KEY EXCHANGE SUCCESSFUL
Both Alice and Bob have the same secret key.
=====

C:\Users\mvluc>
```

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GUI Version :

Code :

```
import tkinter as tk
from tkinter import messagebox

# Diffie-Hellman Calculation
def calculate():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        a = int(alice_entry.get())
        b = int(bob_entry.get())

        if p <= 1 or a <= 0 or b <= 0:
            raise ValueError

        # Public keys
        A = pow(g, a, p)
        B = pow(g, b, p)

        # Shared secret keys
        alice_secret = pow(B, a, p)
        bob_secret = pow(A, b, p)

        # Display results
        alice_public.config(text=str(A))
        bob_public.config(text=str(B))
        alice_secret_label.config(text=str(alice_secret))
        bob_secret_label.config(text=str(bob_secret))

        if alice_secret == bob_secret:
            result.config(
                text="✓ KEY EXCHANGE SUCCESSFUL",
                fg="#00ffcc"
            )
        else:
            result.config(
                text="✗ KEY EXCHANGE FAILED",
                fg="red"
            )

    except ValueError:
        messagebox.showerror(
```

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```
        "Error",
        "Please enter valid positive numbers."
    )

# Clear function
def clear():
    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)
    alice_entry.delete(0, tk.END)
    bob_entry.delete(0, tk.END)

    alice_public.config(text="-")
    bob_public.config(text="-")
    alice_secret_label.config(text="-")
    bob_secret_label.config(text="-")
    result.config(text="READY", fg="#00ffcc")

# Fullscreen toggle
def fullscreen(event=None):
    root.attributes("-fullscreen", not root.attributes("-fullscreen"))

# Exit fullscreen
def exit_fullscreen(event=None):
    root.attributes("-fullscreen", False)

# ----- GUI -----

root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")

# Full screen
root.attributes("-fullscreen", True)

root.configure(bg="#050505")

# Press F11 to toggle fullscreen
root.bind("<F11>", fullscreen)

# Press Escape to exit fullscreen
root.bind("<Escape>", exit_fullscreen)

# Title
```

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```
tk.Label(  
    root,  
    text="DIFFIE-HELLMAN KEY EXCHANGE",  
    font=("Arial", 28, "bold"),  
    bg="#050505",  
    fg="#00ffcc"  
).pack(pady=30)
```

Public Parameters

```
tk.Label(  
    root,  
    text="Public Parameters",  
    font=("Arial", 18, "bold"),  
    bg="#050505",  
    fg="white"  
).pack(pady=10)
```

Prime number

```
tk.Label(  
    root,  
    text="Prime Number (p)",  
    font=("Arial", 13),  
    bg="#050505",  
    fg="white"  
).pack()
```

```
p_entry = tk.Entry(  
    root,  
    font=("Arial", 14),  
    width=20,  
    bg="#111111",  
    fg="#00ffcc",  
    insertbackground="#00ffcc"  
)  
p_entry.pack(pady=5)  
p_entry.insert(0, "23")
```

Primitive root

```
tk.Label(  
    root,  
    text="Primitive Root (g)",  
    font=("Arial", 13),  
    bg="#050505",  
    fg="white"
```

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```
).pack()

g_entry = tk.Entry(
    root,
    font=("Arial", 14),
    width=20,
    bg="#111111",
    fg="#00ffcc",
    insertbackground="#00ffcc"
)
g_entry.pack(pady=5)
g_entry.insert(0, "5")


# Alice and Bob
people = tk.Frame(root, bg="#050505")
people.pack(pady=25)


# Alice
alice_frame = tk.Frame(
    people,
    bg="#111111",
    padx=30,
    pady=20
)
alice_frame.pack(side="left", padx=30)


tk.Label(
    alice_frame,
    text="ALICE",
    font=("Arial", 20, "bold"),
    bg="#111111",
    fg="#00ccff"
).pack(pady=10)


tk.Label(
    alice_frame,
    text="Private Key (a)",
    bg="#111111",
    fg="white",
    font=("Arial", 12)
).pack()


alice_entry = tk.Entry(
    alice_frame,
    font=("Arial", 14),
```

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```
        width=15,
        bg="#222222",
        fg="white",
        insertbackground="white"
    )
    alice_entry.pack(pady=5)
    alice_entry.insert(0, "6")

    tk.Label(
        alice_frame,
        text="Public Key (A)",
        bg="#111111",
        fg="white"
    ).pack(pady=(10, 0))

    alice_public = tk.Label(
        alice_frame,
        text="-",
        font=("Arial", 18, "bold"),
        bg="#111111",
        fg="#00ccff"
    )
    alice_public.pack()

    tk.Label(
        alice_frame,
        text="Shared Secret",
        bg="#111111",
        fg="white"
    ).pack(pady=(10, 0))

    alice_secret_label = tk.Label(
        alice_frame,
        text="-",
        font=("Arial", 18, "bold"),
        bg="#111111",
        fg="#00ffcc"
    )
    alice_secret_label.pack()

# Bob
bob_frame = tk.Frame(
    people,
    bg="#111111",
    padx=30,
    pady=20
```

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```
)  
bob_frame.pack(side="left", padx=30)
```

```
tk.Label(  
    bob_frame,  
    text="BOB",  
    font=("Arial", 20, "bold"),  
    bg="#111111",  
    fg="#cc66ff"  
).pack(pady=10)
```

```
tk.Label(  
    bob_frame,  
    text="Private Key (b)",  
    bg="#111111",  
    fg="white",  
    font=("Arial", 12)  
).pack()
```

```
bob_entry = tk.Entry(  
    bob_frame,  
    font=("Arial", 14),  
    width=15,  
    bg="#222222",  
    fg="white",  
    insertbackground="white"  
)  
bob_entry.pack(pady=5)  
bob_entry.insert(0, "15")
```

```
tk.Label(  
    bob_frame,  
    text="Public Key (B)",  
    bg="#111111",  
    fg="white"  
).pack(pady=(10, 0))
```

```
bob_public = tk.Label(  
    bob_frame,  
    text="-",  
    font=("Arial", 18, "bold"),  
    bg="#111111",  
    fg="#cc66ff"  
)  
bob_public.pack()
```

```
tk.Label(  

```

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```
        bob_frame,
        text="Shared Secret",
        bg="#111111",
        fg="white"
    ).pack(pady=(10, 0))

    bob_secret_label = tk.Label(
        bob_frame,
        text="- ",
        font=("Arial", 18, "bold"),
        bg="#111111",
        fg="#00ffcc"
    )
    bob_secret_label.pack()

# Buttons
buttons = tk.Frame(root, bg="#050505")
buttons.pack(pady=15)

tk.Button(
    buttons,
    text="EXCHANGE KEYS",
    command=calculate,
    font=("Arial", 13, "bold"),
    bg="#006666",
    fg="white",
    activebackground="#00aaaa",
    padx=30,
    pady=10
).pack(side="left", padx=10)

tk.Button(
    buttons,
    text="CLEAR",
    command=clear,
    font=("Arial", 13, "bold"),
    bg="#333333",
    fg="white",
    padx=30,
    pady=10
).pack(side="left", padx=10)

# Result
result = tk.Label(
    root,
```

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```
text="READY",
font=("Arial", 20, "bold"),
bg="#050505",
fg="#00ffcc"
)
result.pack(pady=15)

# Run
root.mainloop()
```

Output :

The screenshot displays a web-based application titled "DIFFIE-HELLMAN KEY EXCHANGE" in red text. The interface is set against a black background. At the top, under the heading "Public Parameters", there are two input fields: "Prime Number (p)" containing the value 23 and "Primitive Root (g)" containing the value 5. Below these, the application is divided into two columns for "ALICE" and "BOB". Alice's section shows a "Private Key (a)" of 6, a "Public Key (A)" of 8, and a "Shared Secret" of 2. Bob's section shows a "Private Key (b)" of 15, a "Public Key (B)" of 19, and a "Shared Secret" of 2. At the bottom, there are two buttons: "EXCHANGE KEYS" (highlighted in red) and "CLEAR" (disabled, shown in grey). Below the buttons, a red message "✓ KEY EXCHANGE SUCCESSFUL" is displayed.

Public Parameters	
Prime Number (p)	23
Primitive Root (g)	5

ALICE	BOB
Private Key (a)	Private Key (b)
6	15
Public Key (A)	Public Key (B)
8	19
Shared Secret	Shared Secret
2	2

EXCHANGE KEYS CLEAR

✓ KEY EXCHANGE SUCCESSFUL

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